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SET: ALPHA

AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

Faculty of Science and Technology

Department of Computer Science

CSC 2210 Object Oriented Analysis and Design (Section: ALL)

Mid Term Examination **Summer 2022-23**

Total Marks: 80 **Moderator:** Mahfujur Rahman **Time:** 2 hours

General Instructions:

1. Answer **MCQs (Part A & Part B)** in the **OMR** and **Diagram Design (Part C and Part D)** in the spaces provided in this question paper.
2. Return the question paper, signed loose sheet (if used), and OMR sheet at the end of the examination.
3. Marks on the right margin indicate full marks.

Specific Instructions:

1. **All the questions are based on OBE program outcomes.**
2. **Attempting Question number 43 is MANDATORY, it will be mapped to CO number one.**
3. Write the **SET NAME (ALPHA/BETA)** in the **OMR** Sheet.

Name: SHUSMITA ANJUM AZIZ		**Proctors Sign:**
ID: 22-47733-2	**Permit No:** 0058031773 73	
Section: K	**Date:** 12.7.23	**Department:** CSE

Part - A Multiple Choice Question

Answer ALL Questions in the OMR **[30 x 1] = 30 Marks**

1. Which of the following is an abstract item?

A. Component B. Active Class
C. Association Class D. Interface Class

2. Which of the following is the objective of a use case diagram?

A. To represent a structural design view B. To express how a system performs its tasks for actors
C. To address the dynamic interaction view of the system D. To establish a common understanding with end users

3. Which one of the following is not part of a static diagram?

A. Association B. Interaction
C. Generalization D. Aggregation

4. Which of the following is not a part of the Class Diagram?

A. Attributes B. Operations
C. Objects D. Stereotype

5. In which of the following phases of RUP the **least** amount of 'Deployment' are done?

A. Inception B. Elaboration
C. Construction D. Transition

6. Reflexive association is

A. Association among objects of two different classes B. Association among objects of the same class
C. Association among objects of two different classes using index D. Association among different subclasses of the same parent

7. Value of an attribute that remains same for all the objects of a class is known as:

A. Default value attribute
B. Derived attribute
C. Slash
D. Class level attribute

8. Which of the following indicates the number of objects that can be related to each other in the association relationship?

A. Association Role
B. Multiplicity
C. Reflexive Association
D. Qualified Association

9. The property of inheritance is denoted by the relationship known as

A. Generalization
B. Realization
C. Aggregation
D. Composition

10. A view that captures the vocabulary of the system is

A. Process view
B. Deployment view
C. Design view
D. Use Case view

11. Which of the following is a behavioral thing that focuses on the lifeline of the object?

A. Interaction
B. Activity
C. State Machine
D. Component

12. Which of the following is a Class?

A. Interface
B. Use Case
C. Generalization
D. Association

13. Which of the following has general elements?

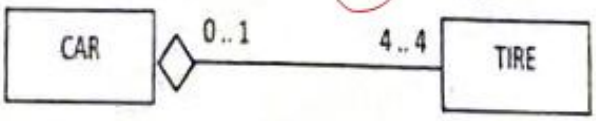
A. Child Class
B. Sub Class
C. Interface
D. Parent Class

14. Qualified association in the class diagram indicates a qualifier to be used for:

A. Inheritance
B. Indexing
C. Interaction
D. Instantiation

15. Which of the following is a prerequisite of interaction?

A. Behavior
B. Identifier
C. Relationship
D. None



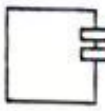
16. Which of the following statements is correct based on the above diagram?

A. A car contains maximum 4 tires and minimum 1 tire.
B. A tire must belong to exactly one car
C. A car contains minimum 4 tires or more
D. A tire may belong to maximum one car or no car at all

Match the notation of **next 4 questions** with the items in the following list:

A. Interface B. Node C. Component D. Active Class E. Artifact

17.



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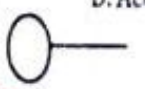

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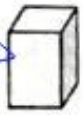


Match the notation of **next 4 questions** with the items in the following list:

A. Dependency B. Generalization C. Interaction D. Aggregation E. Realization

21.

22.

23.

24.

Next 6 questions refer to the following statement:

A **clock** is made of one **machine**, two **arms** and one **dial**. Once assembled the **arms** cannot be used in any other **clock**; but the **engine** and **dial** can be swapped with other **clocks**. **Arms** are connected to the **machine**. The **machine** is driven by one **battery**. There are three types of arms - **Hour, Minute and Second**.

25. The relationship between **clock** and **arm** is:
A. Aggregation
B. Composition
C. Generalization
D. None

26. The relationship between **clock** and **machine** is:
A. Aggregation
B. Generalization
C. Composition
D. None

27. The relationship between **clock** and **dial** is:
A. Aggregation
B. Generalization
C. Composition
D. Association

28. The relationship between **machine** and **battery** is:
A. Aggregation
B. Generalization
C. Composition
D. Association

29. The relationship between **arm** and **hour** is:
A. Association
B. Generalization
C. Realization
D. Aggregation

30. The relationship between **minute** and **hour** is:
A. Aggregation
B. Generalization
C. Composition
D. None

Part - B True/False

Answer ALL Questions in the OMR. Circle 'A' for true and circle 'B' for false **[10x 1] = 10 marks**

31. Relationship among actors is identified using association in a use case diagram.
32. In a use case diagram, a system can be an actor of another system.
33. Generalization is a top-down process.
34. Include dependency relationship in a use case diagram is denoted for optional condition.
35. Objects are templates of classes.
36. Structural things address mostly the dynamic part of the system.
37. Qualified association provides approximately the same functionality as indexes in database.
38. The minimum multiplicity of the association at the whole side can be zero in case of composition.
39. Association Class encapsulates information about relationship.
40. Interaction is prerequisite of behavior.

Part-C **Diagram Design**

Answer ALL Questions. Use the spaces provided in this question paper. You may use **PENCIL** to draw the diagrams. **15 + 10 = 25 Marks**

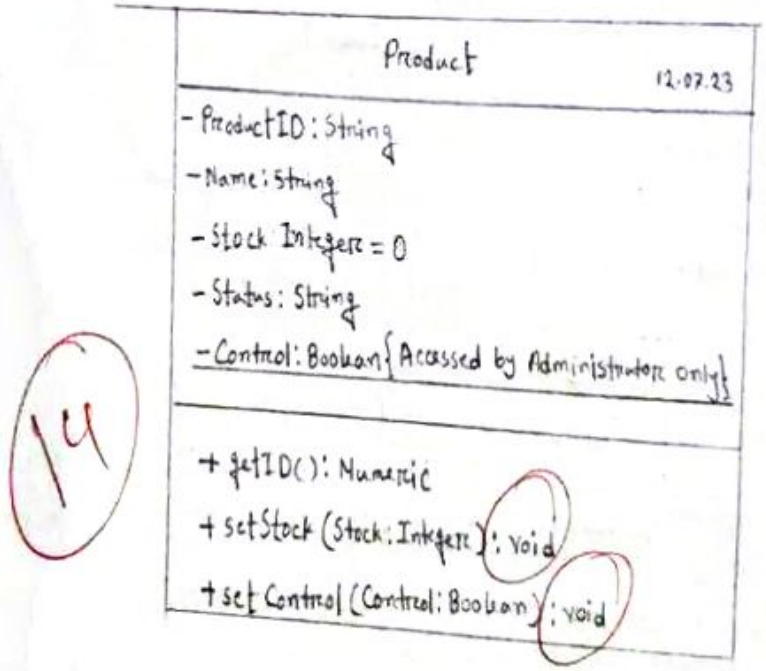
41. **Class Notation:** Draw the Class notation for Class Named **'Product'**. 15 Marks

Private Attributes of the class are:

- **ProductID** (Datatype: String)
- **Name** (Datatype: String)
- **Stock** (Datatype: Integer, default value: zero)
- **Status** (Datatype: String)
- **Control** (Datatype: Boolean, type: static, constraint: Accessed by Administrator only)

Public Operations of this class are:

- **getID** (return datatype: Numeric)
- **setStock** (argument: Stock; datatype: Integer)
- **setControl** (argument: Control, datatype: Boolean)



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3. Case: 03

⇨ Draw the Class notation for Class Named ' **Product** '.

Private Attributes of the class are:

- ProductID (Datatype: String)
- Name (Datatype: String)
- Stock (Datatype: Integer, default value: zero)
- Status (Datatype: String)
- Control (Datatype: Boolean, type: static, constraint: Accessed by Administrator only)

Public Operations of this class are:

- getID (return datatype: Numeric)
- setStock (argument. Stock; datatype: Integer)
- setControl (argument: Control, datatype: Boolean).

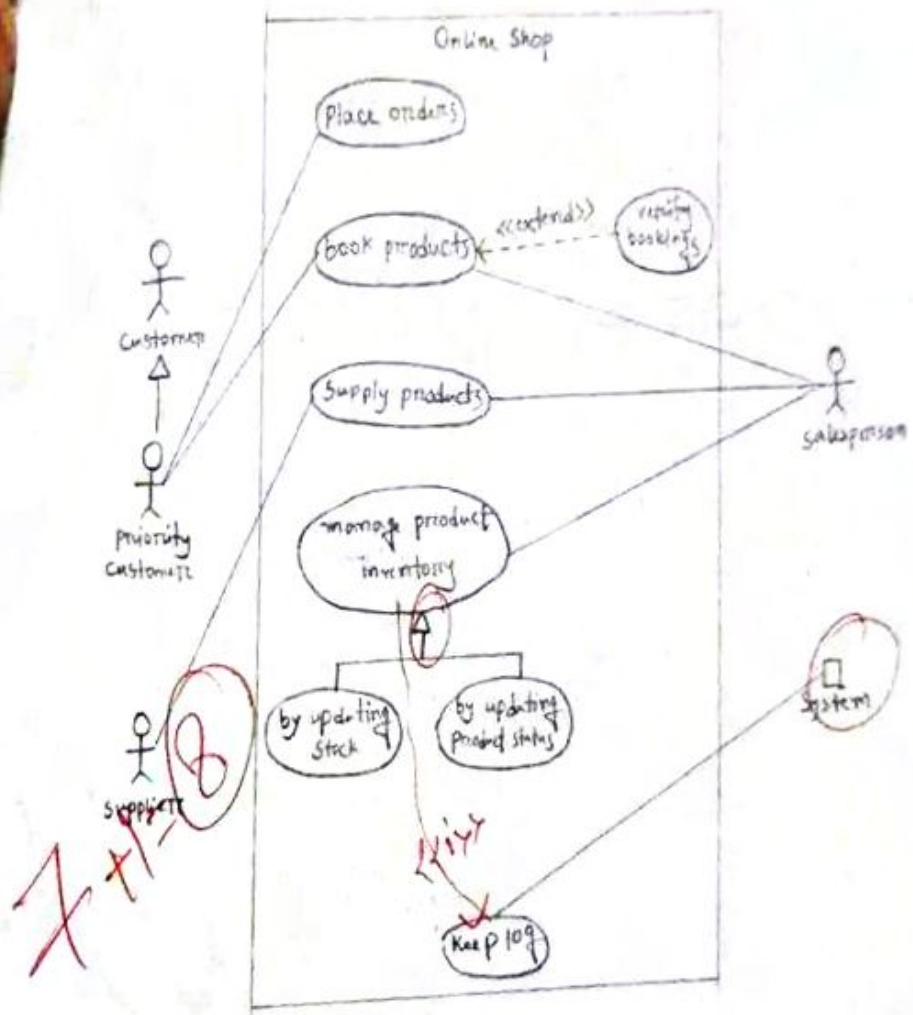
Product
- ProductID: String = blank **- Name: String = blank** **- Stock: Integer = 0** **- Status: String = blank** **- Control: Boolean= none {Accessed by Administrator}**
+ getID (): Numeric **+ setStock (Stock: Integer)** **+ setControl (Control: Boolean)**

42. Draw a **Use Case** diagram for the following requirements: 10 Marks

Customers can place orders. There is a special type of customer called priority customers who can place orders like every other customer, but they can also book products as per their requirements. Booking products are verified by a salesperson before the booking is completed. Sometimes booking verifications are not required for some specific products. Suppliers supply products. A salesperson deals with the supply. A salesperson also manages product inventory. Options of managing product inventory are - updating stock and updating products status. The system must keep a log of all sorts of product management tasks done by the salesperson.

OR

Draw a CRC card for Product based on the information available in cases of question no. 41, 42, 43.



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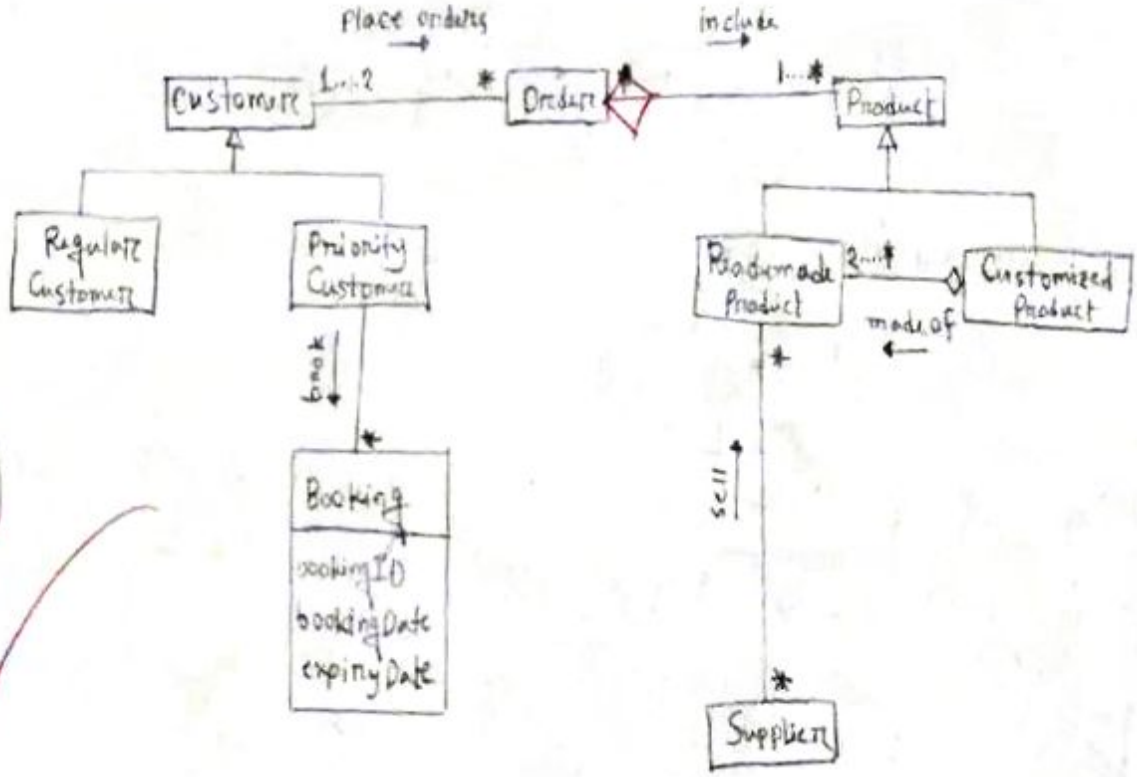
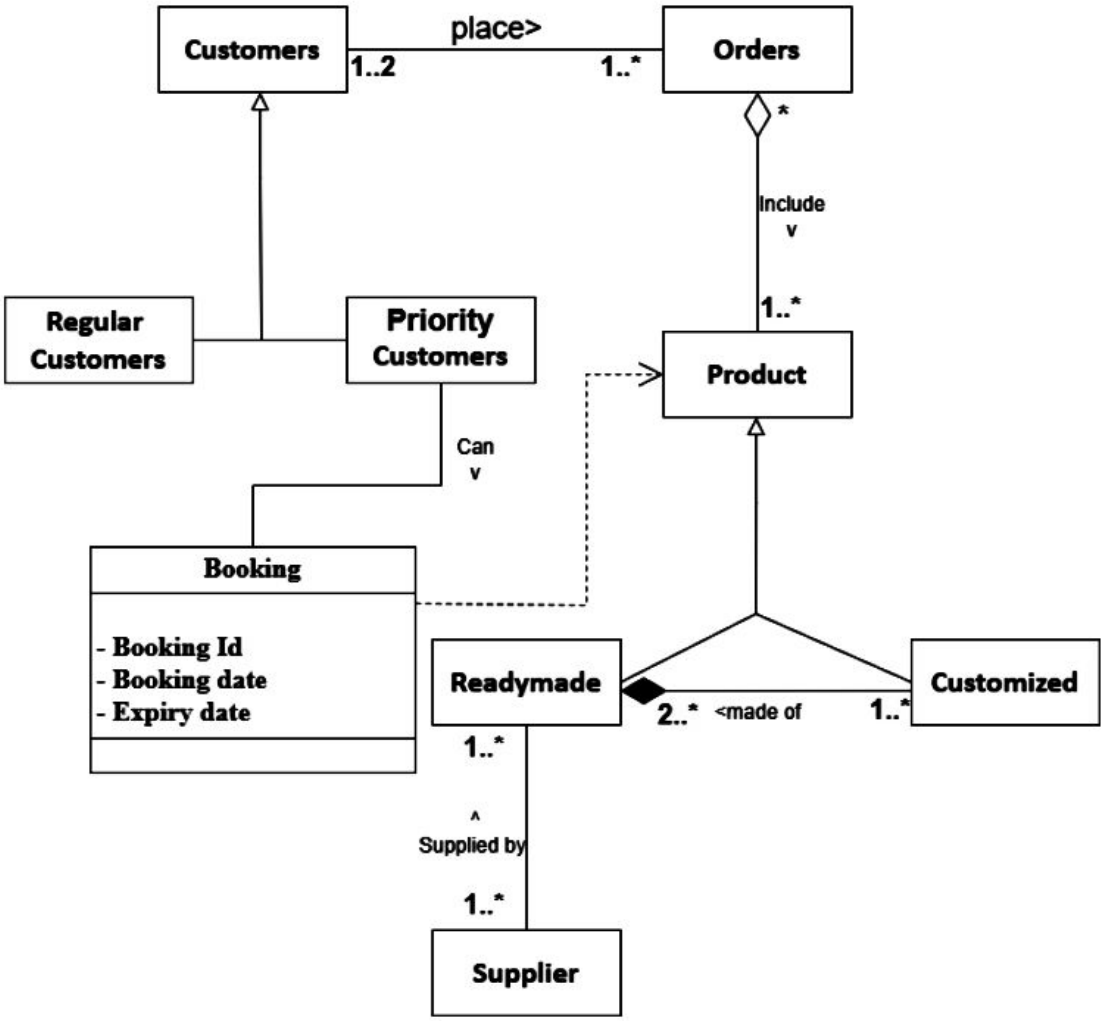
Part-D Diagram Design			15 Marks
This section will be evaluated for the following OBE course outcome.			
Student Name:		**Student ID:**	
CO1: Illustrate the concept of object orientated analysis using UML diagrams.			**Total Marks**
Object Oriented Concept	[5 Marks]		
Diagram Standard	[5 Marks]		
Correctness and Efficiency of the Design	[5 Marks]		

43. Draw a **Class Diagram** for the following order management system: 15 Marks

A system is required to manage customers, and their orders. Customers can place orders for products. One customer can place as many orders as they wish; one order can be placed by minimum 1 or maximum 2 customers. One order must include at least one product. One product may be included in many orders. The organization sells two types of products – readymade and customized. A readymade product is sold as it is bought from the supplier. A customized product is made of at least 2 readymade products. One readymade product can be bought from many suppliers. One supplier can sell many readymade products. The organization has two types of customers – regular and priority. Priority customers can also book products. Every booking is identified by a booking ID. There are also booking date and expiry date attributes for every booking.



Case: 04

- A system is required to manage customers, and their orders. **Customers** can place **orders** for products. One customer can place as many orders as they wish; one order can be placed by minimum 1 or maximum 2 customers. One order must include at least one **product**. One product may be included in many orders. The organization sells two types of products - **readymade** and **customized**. A readymade product is sold as it is bought from the **supplier**. A customized product is made of at least 2 readymade products. One readymade product can be bought from many suppliers. One supplier can sell many readymade products. The organization has two types of customers-**regular** and **priority**. Priority customers can also book products. Every booking is identified by a booking ID. There are also booking date and expiry date attributes for every booking.